Critical Site Percolation

A conditional theorem, and the two statements that remain

3 September 2026

Draft, work in progress. Two statements in this note are not yet proved. They are printed in this colour wherever they appear, and they are collected in Section 5. Everything else stated here is proved and machine-checked in Lean, with no unproved step and no axioms beyond the three that Lean itself uses.

Abstract

Keep each site of the integer lattice with probability p, independently of the others, and ask whether the origin belongs to an infinite cluster of kept sites. The chance of that is zero for small p and positive for large p, and the value it takes at the threshold between the two is not known in three or more dimensions. This note reduces that value to two finite statements. The first is a single inequality about connection probabilities, which covers bond percolation and site percolation at once. The second is a reduction of the lattice to a slab of finite width. From these two, and from nothing else, the chance of an infinite cluster at the threshold is zero.

1 The question

In site percolation each site of the lattice is kept with probability p and discarded otherwise, independently of every other site. Two kept sites belong to the same cluster when some path of neighbouring kept sites joins them. A discarded site belongs to no cluster at all, not even its own. Write $\theta(p)$ for the chance that the origin belongs to an infinite cluster.

Raising p keeps more sites and can only enlarge clusters, so θ increases with p. There is therefore a threshold p_c with $\theta(p) = 0$ below it and $\theta(p) > 0$ above it. Those two facts are immediate from the definition, and together they determine θ at every parameter except one. At the threshold itself an infinite cluster is neither forced nor forbidden, and whether $\theta(p_c) = 0$ is the question.

In two dimensions the answer is yes. For bond percolation, where edges rather than sites are kept, it is due to Harris and Kesten. For site percolation it is due to Russo, whose argument follows Kesten's determination of the bond threshold on the square lattice. In three and more dimensions the answer is yes for bond percolation, by a recent machine-checked proof that settled the dimensions three to ten. For site percolation in three or more dimensions it is not known, and that is the case treated here.

2 One inequality for both models

The finite input concerns a structure slightly more general than a graph. Take a finite set of sites and a finite set of labels, and attach to each label some collection of sites. Keep each label with its own probability, independently. Two sites are connected when a chain of kept labels joins them, consecutive labels in the chain sharing a site. Connection is reflexive, so every site is connected to itself.

Two familiar models are instances. If every label is attached to exactly two sites, the labels are edges and this is bond percolation. If instead each site carries one label of its own, attached to that site and to one extra point for each of its edges, then keeping a label means keeping a site, and this is site percolation. The second description is proved below.

Definition 2.1 (the finite inequality)**.** For a nonempty set A of sites and sites o and b,

$$\mathbb{P}(o \leftrightarrow b \text{ and } o \leftrightarrow A) \;\geq\; \mathbb{P}(o \leftrightarrow A) \min_{a \in A} \mathbb{P}(a \leftrightarrow b).$$

Read from right to left, it says that reaching the set A and then reaching b from the worst point of A is no better than reaching b directly. For labels attached to two sites, and after dropping the second event on the left, this is the inequality Kozma and Nitzan conjectured, which is proved and machine-checked for that case.

Definition 2.1 in the stated generality is not yet proved.

3 Site percolation is a model of labels

The description of site percolation promised above is exact, not an analogy.

Theorem 3.1. *Let a finite graph be given, with each site kept with its own probability. Give each site one label, attached to that site and to one further point for each edge at it, and keep that label exactly when the site is kept. Then two distinct sites are connected in the model of labels precisely when they are connected by a path of kept sites in the graph.*

Proof. Since the labels are the sites, the two models have the same sample space and the same probabilities, so only the connection relation has to be matched. A kept label attached to a site reaches, besides that site, one further point for each edge at it, and the only other label reaching that point is the one of the site at the other end of the edge. A chain of kept labels from one site to another therefore alternates between sites and the points that stand for the edges between them, and reading off the sites gives a path of kept sites. Conversely a path of kept sites gives such a chain. □

Corollary 3.2. *The inequality of Definition 2.1 yields the same inequality for site percolation on every finite graph, with no site required to be kept in advance.*

Why labels rather than the sites they reach

Arguments of this kind proceed by exposing the labels that touch a set and recording what they reach outside it. For graphs, recording the set of outside sites reached is enough. For labels attached to more than two sites it is not, and this is the one place where the general case genuinely differs from the graph case. Two distinct labels can reach exactly the same outside sites, so passing from a set of labels to the sites it reaches loses information: the record of a union is the union of the records, but the record of an intersection can be strictly larger than the intersection of the records. A two-label example witnessing this is included in the formalization. The arguments below therefore keep the labels themselves rather than the sites they reach.

4 From the inequality to the lattice

The passage from a finite inequality to the lattice is a renormalization: the lattice is divided into blocks, a block is declared occupied when a suitable connection is present inside it, and the occupied blocks are compared with ordinary percolation on a coarser lattice. Carrying that comparison out requires connecting one block to the next, and the classical way to do it is to keep a few extra sites beyond those the construction has already used. Grimmett and Marstrand do exactly this and prove that percolation at a parameter forces percolation in a slab of finite width at any slightly larger parameter. They also record that they cannot remove the excess, and that this is what stops their argument from settling the value at the threshold.

The inequality of Definition 2.1 is meant to replace those extra sites. What it should deliver is the following.

Definition 4.1 (the slab reduction)**.** If the chance of an infinite cluster at a parameter strictly inside the unit interval is positive, then percolation occurs in a slab of finite width, at the same parameter or at a smaller one.

Definition 4.1 is not yet proved.

Everything from here to the conclusion is proved. Two routes are available, and they differ in which form of the slab reduction they consume.

Theorem 4.2 (the self-contained route)**.** *Suppose that percolation at a parameter strictly inside the unit interval forces percolation in a slab of finite width at some strictly smaller parameter. Then the chance of an infinite cluster at the threshold is zero.*

Proof. A slab sits inside the lattice, and percolation is monotone under enlarging the vertex set, so percolation in the slab at a parameter gives percolation in the lattice at that parameter. Suppose the chance of an infinite cluster at the threshold were positive. The threshold lies strictly inside the unit interval, so the hypothesis applies and produces a smaller parameter at which the lattice percolates. That contradicts the threshold being the infimum of the parameters at which it does. □

This route assumes nothing else. No theorem about percolation at a critical point enters it, and the two facts that the threshold is neither zero nor one are themselves proved, the first by counting paths and the second by comparing site percolation with bond percolation.

Theorem 4.3 (the route through a half-space)**.** *Suppose that percolation at a parameter forces percolation in a slab of finite width at the same parameter. Suppose in addition that the chance of an infinite cluster in a half-space vanishes at the threshold of the half-space, and that in three or more dimensions this threshold agrees with the one of the lattice. Then the chance of an infinite cluster at the threshold is zero.*

The two additional suppositions are published theorems. The first is due to Barsky, Grimmett and Newman, who prove it for bond percolation and for site percolation alike. The second is due to Grimmett and Marstrand, whose paper works throughout with site percolation. Both are needed, since the first alone places the statement at the threshold of the half-space rather than at the threshold of the lattice.

5 What remains

Two statements are not yet proved, and only these two.

1. The inequality of Definition 2.1, for labels attached to any number of sites. For labels attached to two sites it is proved.
2. The slab reduction of Definition 4.1.

Both appear in the formalization as named hypotheses, so every statement that uses them displays them and nothing can quietly depend on them. Given the first, the second is what the whole construction is for, and given both, the conclusion follows by Theorem 4.2 with no further input.

6 The formalization

The definitions, the theorems and the corollary above are formalized in Lean 4 with Mathlib. The model, the connection events, the chance of an infinite cluster and the threshold are all definitions in the files, and the threshold is proved to lie in the unit interval, so the conclusion is an equation between two real numbers rather than a statement that could hold for want of content.

Also proved there, and used above: that conditioning on the whole cluster of a set factorizes exactly; that two disjoint clusters may be factorized in either order; that any probability expands as a finite sum over the possible clusters; that the chance of an infinite cluster is monotone both in the parameter and in the underlying graph; that it vanishes below the threshold; that the threshold is neither zero nor one; that events depending on finitely many sites have probabilities varying Lipschitz continuously in the parameter; and that an exploration whose every step succeeds with probability at least a fixed amount, whatever has happened before, reaches any upward-closed target at least as often as an independent sequence would.

Every one of those compiles with no unproved step, and each is accepted using only the three axioms that Lean itself provides.

References

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